



NANYANG
TECHNOLOGICAL
UNIVERSITY
SINGAPORE

SC3021 – Data Science Fundamentals

Final Presentation Write-Up

Group 3, SDAC

Shadid Zarraf Rahman (U2420740B)

Steven Octavianus Rahmat (U2420184J)

Wee Keng Kit, Wayne (U2420329C)

2025 – 2026 Semester 2



Problem Statement and Goal

"How do different learning methods, lifestyle factors, and AI-assisted learning behaviours influence students' academic performance, and what actionable strategies can be recommended to improve outcomes?"

- **Question:** What factors most influence students' final grade?
- **Goal:** Analyze relationships and build prediction models for academic performance
- **Target variable:** final score (continuous academic score)

Data Sources

Datasets Used:

- **DS1:** Academic performance (main dataset)
- ~~**DS2:** Lifestyle and Daily Habit (Discarded – due to low correlation)~~
- **DS3:** Social media behaviour
- **DS4:** Psychological wellbeing

Challenge:

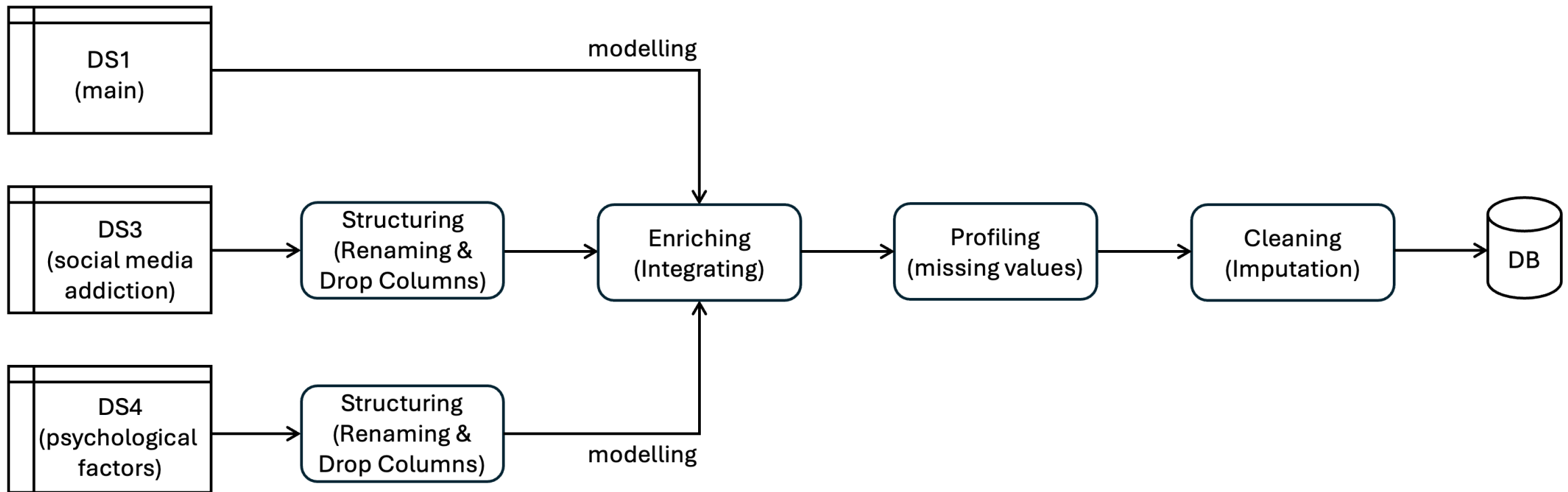
- Datasets come from different students → **cannot merge directly**

Features of Main Dataset

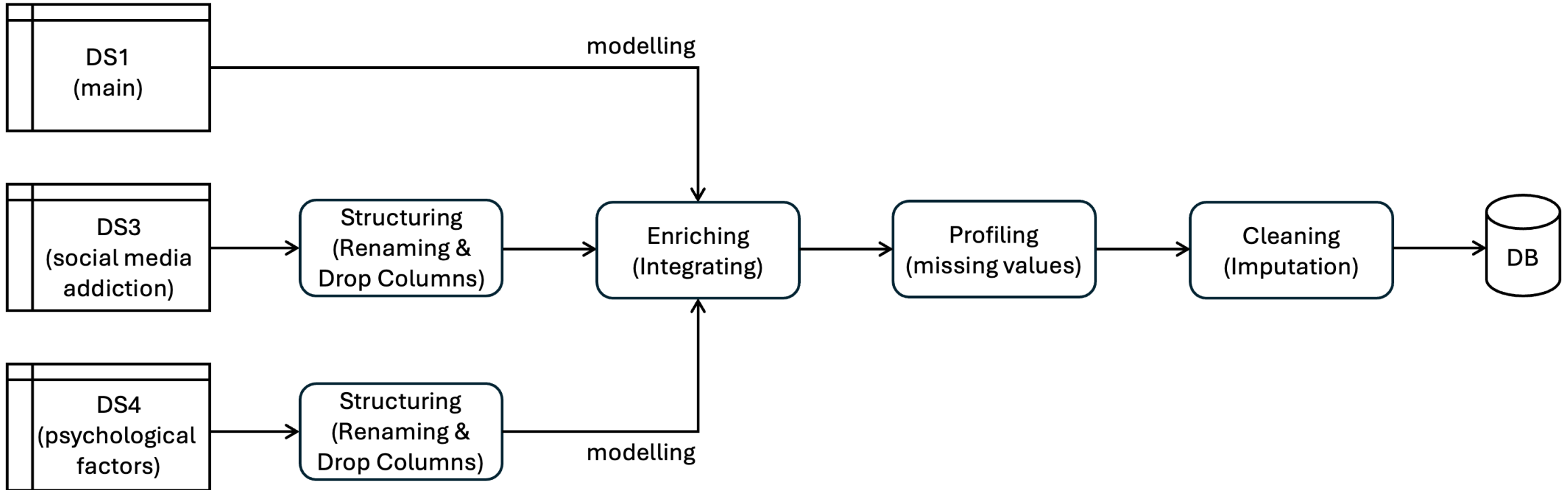
- **Student Demographics:** student id, age, gender, grade level.
- **Study Behaviour and Learning Habits:** study hours per day, study consistency index, tutoring hours, class participation score.
- **AI Usage in Learning:** uses ai, ai usage time, ai tools used, ai usage purpose, ai dependency score, ai generated content percentage, ai prompts per week, ai ethics score.
- **Lifestyle and Daily Behaviour:** sleep hours, social media hours.
- **Academic Performance Indicators:** last exam score, assignment scores average, attendance percentage, concept understanding score, improvement rate, final score.

How we merge the data?

- **Solution:** Model-based **feature transfer**
→ use shared variables (age, sleep, social media)



How we merge the data?



- **Feature Enrichment:**

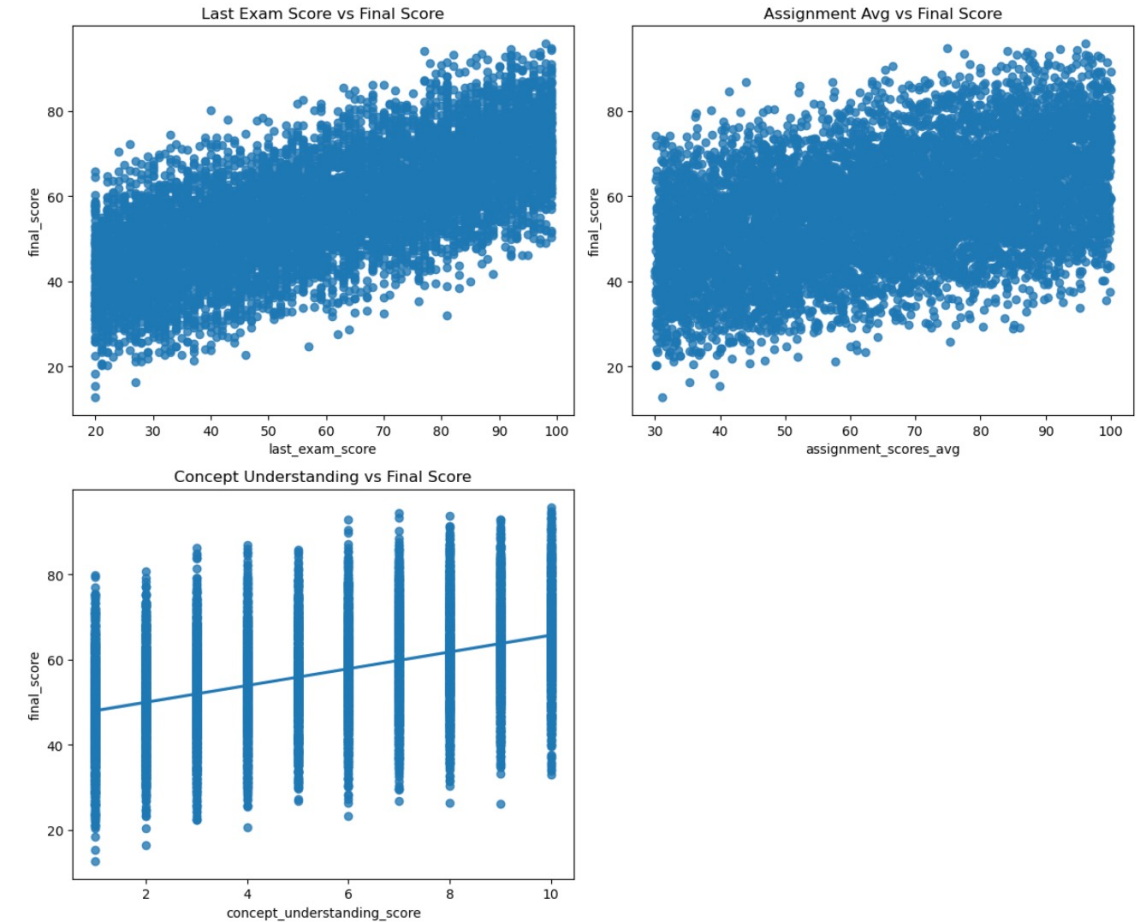
- DS3 → predicted social media addicted score
- DS4 → predicted wellbeing, predicted burnout risk

- **Result:**

- One enriched dataset combining academic, behavioural, and psychological factors

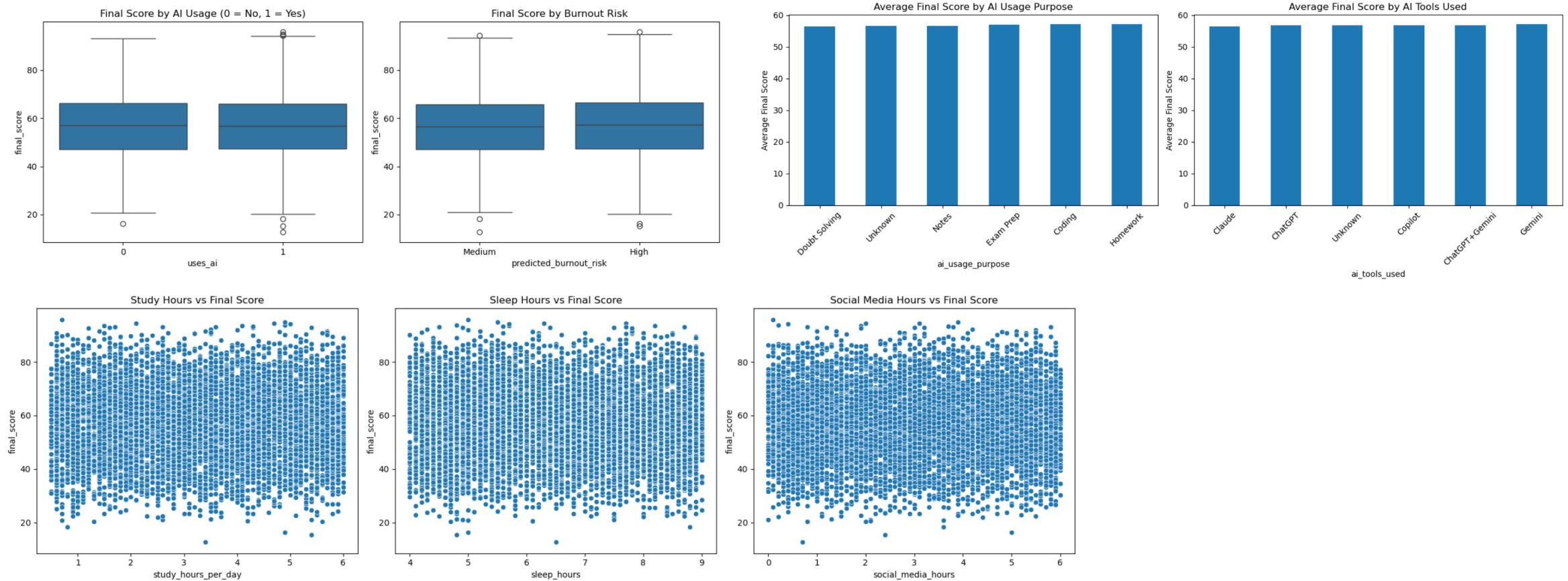
What drives student's performance?

- **Strong positive relationships:** last exam score, assignment scores avg, concept understanding score.



What drives student's performance?

- **Weak / no clear relationship:** Study hours, AI usage, Lifestyle factors



We also perform statistical testing...

- **Statistical testing confirms:**
 - Prior performance → significant ($p \approx 0$)
 - Study hours → not significant
 - AI usage → no difference
 - Burnout → weak / borderline effect

Modelling

- **Feature Group Performance (R^2):**

- **Prior Performance → 0.68**
- Academic Behaviour → 0.19
- AI Usage → ~0.00
- Lifestyle → ~0.00
- Psychological Factors → ~0.00

- **Combined Model:**

- **Linear Regression → 0.87**
- XGBoost → 0.86
- Random Forest → 0.86

Key Takeaway

Key Takeaways:

- Academic performance is mainly driven by:
 - Prior achievement
 - Conceptual understanding
- Study hours alone do not guarantee better results
- AI usage has **no direct impact**
- Burnout has a **small but noticeable effect**

Limitation and Challenges

Limitation:

- Datasets from different populations → required feature transfer
- Some variables are **predicted, not observed**
- Limited ability to infer causality

Final Message:

- Focus on **learning quality, not quantity**
- Early performance is key to long-term success